

ALLISON WANG

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PERSONAL STATEMENT

Fourth year mechanical engineering student at the University of California, Berkeley, looking to pursue a career in aerospace hardware. Currently have three years of project experience relating to rocketry as a part of payloads design project at UC Berkeley's Space Technologies and Rocketry team. Previously developed avionics test and simulation tools, including C/C++ software for bus signal monitoring and aircraft-system integration and testing. Passionate about the intersection between aerospace and fundamental engineering principles to develop mechatronics for the future of air mobility.

EDUCATION

University of California, Berkeley | *B.S. in Mechanical Engineering, 3.82 GPA*

Expected May 2027

Relevant Coursework: Introduction to Composite Materials, Experimentation and Measurements, Heat Transfer, Aerodynamics

RESEARCH AND PROJECT EXPERIENCE

Lawrence Berkeley National Laboratory (LBNL) Inorganic Nanostructures Facility | *Robotics Intern* Sep 2026 – Present

- Design an industrial automation robot used to perform repetitive tasks within the Molecular Foundry - Inorganic Nanostructures facility and document operation instructions within a standard operating procedure
- Research and develop an end effector to push and pull at many different angles, open and close vial caps of varying sizes, dispense powders and liquids into other containers

Composites Knee Prosthesis | *Project Member*

Jan 2026 – May 2026

- Researched and exercised Hertz contact theory to validate hand calculations for complex knee loading conditions
- Prototyped knee replacement in SOLIDWORKS, creating detailed, optimized structure for lightweight structural solutions
- Modeled various loading conditions in ANSYS FEA and optimized composite stackup using a genetic algorithm in Python, verifying structural integrity of various 6-layer composite fiber-matrix combinations
- Conducted 10+ Instron small-scale prototype tests, validating loading conditions of the knee prosthetic

Dog Pill Dispenser | *Project Member*

Jan 2026 – May 2026

- Designed, prototyped and tested an automated dog pill dispenser that precisely injects coating around a standard pill to enhance flavor and dog obliging to take the pill, reducing the hassle and manual labor of current dog pill coating solutions
- Surveyed 20+ dog owners on current dog pill coating solutions, formulated Pugh matrix to narrow down prototypes
- Performed hand calculations of maximum fluid coating capacity, conducted FEA to analyze load bearing of actuating mechanism of syringe
- Outlined state diagrams within dog pill coating process and set up embedded code for sensors, motors and actuators, ensuring MVP functionality

Multiphase Thermofluidics and Surface Nanoengineering | *Undergraduate Researcher*

Aug 2025 – Dec 2025

- Design and implement a test procedure to apply superhydrophobic coatings on wafer surfaces
- Organize chemicals in wet lab environments to ensure proper labeling, storage, and compliance with safety protocols

IoT Drumstick Tutor | *Project Manager*

Jan 2025 – May 2025

- Developed mechanical actuators, firmware code and electronic schematics for ESP-32 microcontroller and live Flask data visualizations to track drumstick position, tempo, and angle in real time
- Tuned a PID controller to achieve precise motorized drumstick angle matching, demonstrating iterative embedded software development and controls optimization

WORK EXPERIENCE

The Boeing Company | *Lab Systems Engineering Intern*

May 2026 – August 2026

- Engineered a C/C++ software GUI application to manage 30+ discrete signals, utilizing shared memory for synchronized read/write buffers and circular queue implementation
- Modernized desktop simulation software by building a C/C++ monitoring window for ARINC 429 bus protocol to streamline signal conversion
- Executed test procedures on VMEbus chassis by analyzing bus addresses and probing test points to verify card initialization
- Improved configuration management workflow by updating drawing versions and converting obsolete documents into CAD, reducing version inconsistencies for 300+ documents

Statistics and Data Science for Engineers | *Reader*

Jan 2025 – May 2025

- Grade weekly homework for 180 students regarding data science fundamentals, optimization, and probability
- Respond to student inquiries, regrade requests, and set up rubrics for machine learning capstone projects

Wisk Aero | Product Quality Engineering Intern

Jun 2025 – Aug 2025

- Spearheaded a dashboard of Non-conformance Reports (NCRs) to categorize 1200+ manufacturing defects, summarize design changes, and identify potential proactive measures to improve production flows
- Executed a Python Natural Language Processing (NLP) pipeline to sort defect reports by key terms, expanding defect coverage by 140%
- Led a cross-functional initiative with 15+ engineers to publish standardized firmware logging processes of 10+ components, enhancing traceability within manufacturing system
- Defined a standardized drawing review process for quality engineers to define exit criteria across battery, composites and propulsion engineering teams

CITRIS Workforce Innovation Program | Embedded Systems Intern

Mar 2024 - July 2024

- Refined PCB designs for manufacturability in KiCAD, integrating IMUs, thermal cameras, and Bluetooth sensors
- Trained and designed Tensorflow algorithms for real-time wildfire hazard detection on an IoT-powered drone and power grid safety system

iD Tech Camps | Machine Learning and Robotics Instructor

Jun 2023 – Sep 2023

- Introduced 80+ elementary students to combat robotics, leading daily 8-hour lessons on strategy, coding and product design
- Personalized learning for 20+ students learning machine learning, teaching OpenCV applications to help students make motion-controlled online games

EXTRACURRICULARS

CalSol (UC Berkeley Formula Solar Racing Team) | Dynamics Engineer

Jan 2026 - Present

- Coded a MATLAB program to tune pedal ratios, optimizing brake requirements and successfully reducing forward bias
- Designed and manufactured a custom caliper mount to reduce stress concentrations and increase safety factors by 200%
- Machined and tested 10+ carbon fiber A-arms by executing destructive Instron tests and injecting epoxy to analyze structural integrity and material limits

Space Technologies and Rocketry | Payloads Lead Engineer & Outreach Lead

Jan 2024 - June 2026

- Utilized controls theory principles and the extended Kalman Filter theory to tune parameters for liquid engine blowdown system, predicting optimal fuel duty cycles based on tank pressure
- Led design, test, and prototyping of fluid circulation life support system using SOLIDWORKS and water-jet
- Applied stress analysis and fluid mechanics principles to optimize high-altitude yeast payloads flown to ~30,000 ft

Concrete Canoe | Computational Fluid Modeling Member

Aug 2024 – Dec 2024

- Utilized Fusion 360 to design canoe hull, simulate 3+ flow conditions using computational fluid dynamics
- Defined loading conditions on concrete canoe design using simple beam theory
- Optimized curvature of concrete canoe cross section using MATLAB and image processing, reducing turbulence towards the edges of the canoe

Pioneers in Engineering | Game Design Lead

Jun 2023 – Present

- Constructed a custom 18 x 36 foot catapult for Lawrence Berkeley National Laboratory's STEM outreach program, recruiting over 30 students for a low-cost robotics competition
- Modeled competition playing fields using Fusion 360 for semesterly robotics events, engaging 50+ underserved high school teams and promoting collaboration, teamwork, and accessible STEM education

Society of Women Engineers (SWE) | Evening with Industry and Overnight Host Program Committee

Jun 2023 – Present

- Mentor incoming freshmen in preparation for college life at UC Berkeley, providing STEM advice and resources
- Organized annual career fair for 80+ students, managing catering, pamphlets and venue
- Outreached to 10+ companies to table for club fair for incoming students and designed pamphlets to support recruitment
- Designed and coded host program resources page, allowing easy navigation of resources, maps and program schedule

RELEVANT SKILL SETS

- **Mechanical Design:** Computer-Aided Design (CAD) and Computer-Aided Machining (CAM) in Solidworks, Fusion 360, Adobe Illustrator, Figma
- **Electronics:** Circuit Design in Multisim, KiCAD, Breadboarding, Soldering, Arduino
- **Languages:** Native English, Basic Mandarin
- **Communication:** Powerpoint, Pugh Charts, Engineering-Integration Collaboration, Requirements Validation
- **Lab:** Multimeter, Oscilloscope, LabVIEW, MATLAB, Java, Python
- **Software:** Simulink, C/C++ Firmware programming
- **Hardware:** 3D Printing (Prusa), Laser Cutting, Machining (Mill, Lathe, Drill Press, Chopsaw, Bandsaw), Geometric Dimensioning and Tolerancing (GD&T)